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BRIEF COMMUNICATION

Indigomycin, a new pyridinetrione from *Streptomyces albus* J1074 harboring the indigoidine biosynthetic gene cluster

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Abstract

Heterologous expression of the indigoidine biosynthetic gene cluster from *Streptomyces laurentii* ATCC 31255 in *Streptomyces albus* J1074 resulted in the production of the known blue pigment indigoidine, along with a structurally novel pyridinetrione derivative designated indigomycin (**1**). The structure of indigomycin was determined through comprehensive 1D and 2D NMR spectroscopy and X-ray crystallographic analysis. Antimicrobial evaluation revealed that indigomycin (**1**) exhibits modest activity against methicillin-resistant *Staphylococcus aureus* ATCC 43300, with a minimum inhibitory concentration (MIC) of 32 μg mL^{-1}. Biosynthetically, the pathway is proposed to be initiated by a single multi-domain NRPS (IgoA) that activates L-glutamine and directs the divergent assembly of both indigoidine and indigomycin from a common dehydroglutaminyl intermediate. This work not only enriches the structural diversity of pyridinetrione-type natural products but also presents an efficient biosynthetic strategy for sustainable production of indigoidine-based pigments.

Indigoidine is a bipyridyl pigment with a longstanding use in the dye industry, originally discovered in microorganisms in the 1960s [1, 2]. The biosynthetic gene clusters (BGCs) responsible for its production are widely distributed across diverse bacterial lineages, notably among Actinobacteria and Proteobacteria [3–5]. Its biosynthesis is centrally catalyzed by a single-module non-ribosomal peptide synthetase (NRPS). This NRPS initiates the process by recognizing L-glutamine as the starter substrate via its adenylation (A) domain, which incorporates an integrated oxidation module (Ox) [4, 6]. The activated substrate is then shuttled by the peptidyl carrier protein (PCP) and ultimately released by the thioesterase (TE) domain. Recent genetic engineering of indigoidine synthetase genes has enabled efficient heterologous production of this water-insoluble blue pigment [7–9], while also facilitating its application as a cross-kingdom reporter system [10] and enabling the generation of blue roses [11]. Notably, a recurring genomic association between indigoidine synthetase and glycosyltransferase genes has proven valuable in natural product discovery, leading to the identification of compounds such as the water-soluble blue pigment indochrome and the *C*-nucleoside antibiotic minimycin [4, 6]. This conserved genetic linkage highlights the promise of leveraging indigoidine synthetases as genomic markers for the discovery of novel bioactive molecules.

During our ongoing genomic mining efforts [12–14], a putative indigoidine BGC was identified in *Streptomyces laurentii* ATCC 31255, which harbors a core indigoidine synthetase gene. We employed the CAPTURE system, a Cas12a-assisted precise targeted cloning using in vivo Cre-*lox* recombination [12], and successfully cloned a 26-kb region designated as *igo*. Heterologous expression of the *igo* in *Streptomyces albus* J1074 led to the production of both the known blue pigment indigoidine and a novel metabolite, named indigomycin (**1**). Herein, we described the cloning and heterologous expression of the gene cluster

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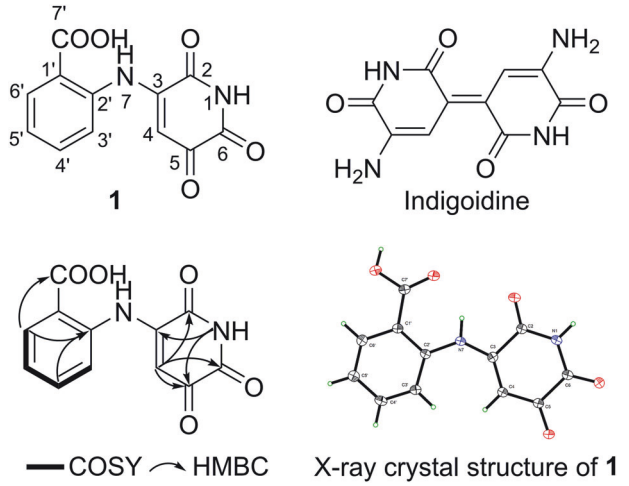

Fig. 1 Chemical structures of compounds indigomycin (**1**) and indigoidine. Selected COSY and HMBC correlations and the X-ray crystal structure of **1** are also shown

igo, and the isolation, structural elucidation, and antibacterial activity of compound **1**.

The 26-kb *igo* BGC was successfully cloned from *S. laurentii* ATCC 31255 using the CAPTURE system and heterologously expressed in *S. albus* J1074, a host strain well-suited for secondary metabolite production. After 30 L fermentation, the crude extract was fractionated to afford two major metabolites. These were identified as the known blue pigment indigoidine (4.0 mg) [6], as well as a structurally novel pyridinetrione analogue (3.0 mg), designated indigomycin (**1**) (Fig. 1). High-resolution electrospray ionization mass spectroscopy (HRESIMS) established the molecular formula of indigomycin (**1**) as $C_{12}H_8N_2O_5$ (*m/z* 261.0565 $[M+H]^+$, calcd 261.0506), corresponding to 10 degrees of unsaturation. UV-vis spectroscopy showed a maximum absorption at 422 nm, indicative of an extended conjugated system. The ^{1}H NMR spectrum (Table 1) displayed signals for four aromatic protons δ_H 7.62 (d, $J = 8.1$ Hz, H-3′), 7.67 (ddd, $J = 8.1$, 7.9, 1.3 Hz, H-4′), 7.28 (ddd, $J = 7.9$, 7.8, 0.9 Hz, H-5′), 8.04 (dd, $J = 7.8$, 1.3 Hz, H-6′), an olefinic singlet at δ_H 6.44 (s, H-4), and two exchangeable protons at δ_H 11.15 and 12.27. The ^{13}C NMR spectrum revealed seven aromatic and four carbonyl carbons. The presence of a characteristic 1,2-disubstituted benzene ring in **1** was established by the H-3′/H-4′/H-5′/H-6′ COSY spin system (Fig. 1). Key HMBC correlations from H-4 to C-2, C-3, C-5, and C-6, together with correlations from 1-NH to C-3 and C-5, supported the assignment of a 3-amino-2,5,6-pyridinetrione unit. Additionally, HMBC correlations from H-5′ to C-3′ and C-1′, and from H-6′ to C-7′, C-4′, and C-2′, further indicated the presence of an anthranilic acid moiety [15]. Finally, the structure of **1** was unambiguously determined by X-ray crystallographic analysis (Fig. 1).

Table 1 ^{1}H and ^{13}C NMR data for indigomycin (1) (TMS, δ in ppm)

Position	δ_H, multi (J in Hz)	δ_C, type
1	12.27, s	
2		157.2, C
3		143.1, C
4	6.44, s	103.2, CH
5		174.8, C
6		161.4, C
7	11.15, s	
1′		121.3, C
2′		138.9, C
3′	7.62, d, 8.1	121.3, CH
4′	7.67, ddd, 8.1, 7.9, 1.3	133.8, CH
5′	7.28, ddd, 7.9, 7.8, 0.9	124.5, CH
6′	8.04, dd, 7.8, 1.3	131.9, CH
7′		168.5, C

^{1}H and ^{13}C NMR were recorded at 600 and 150 MHz in DMSO-d_6, respectively

Evaluation of the minimal inhibition concentration (MIC) revealed that compound **1** exhibited no significant antibacterial activity against any of the five tested strains (Table 2): *Staphylococcus aureus* ATCC 29213, *Acinetobacter baumannii* ATCC 19606, *Bacillus subtilis* SCSIO BS01, *Pseudomonas aeruginosa* ATCC 9027, and methicillin resistant *S. aureus* ATCC 43300.

We propose that a single-module, multi-domain NRPS, IgoA, which contains A-Ox-PCP-TE domains, orchestrates a bifurcating pathway from a common L-glutamine precursor to direct the divergent biosynthesis of indigomycin and the blue pigment indigoidine (Fig. 2). The pathway begins with the A domain selectively activating and loading L-glutamine onto the adjacent PCP domain, forming an L-glutaminyl-*S*-PCP thioester. This intermediate is subsequently oxidized by the Ox domain, yielding a dehydroglutaminyl-*S*-PCP species. This activated intermediate represents a critical branch point. For indigoidine biosynthesis, the dehydroglutaminyl-*S*-PCP is channelled into a cyclization reaction catalyzed by the TE domain to form 5-amino-3*H*-pyridine-2,6-dione (**M1**). **M1** then undergoes spontaneous, non-enzymatic dimerization to yield the blue pigment indigoidine [4]. Alternatively, for indigomycin production, the same dehydroglutaminyl-*S*-PCP intermediate is hydrolyzed, generating 5-amino-2,5-dioxopentyl-*S*-PCP. The linear keto-acid thioester then undergoes intramolecular cyclization to form piperidinetrione (**M2**). It is also possible that **M1** undergoes direct imine hydrolysis to yield **M2**. We postulate that **M2** serves as an essential and putative key precursor for the assembly of the indigomycin (**1**) scaffold. The transformation of **M2** into indigomycin is hypothesized to involve multiple enzymatic steps,

Table 2 Antibacterial activity of 1

MIC (μg mL^{-1})				
S. aureus ATCC 29213	*A. baumannii* ATCC 19606	*B. subtilis* SCSIO BS01	*P. aeruginosa* ATCC 9027	methicillin resistant *S. aureus* ATCC 43300
>64	>64	>64	64	32

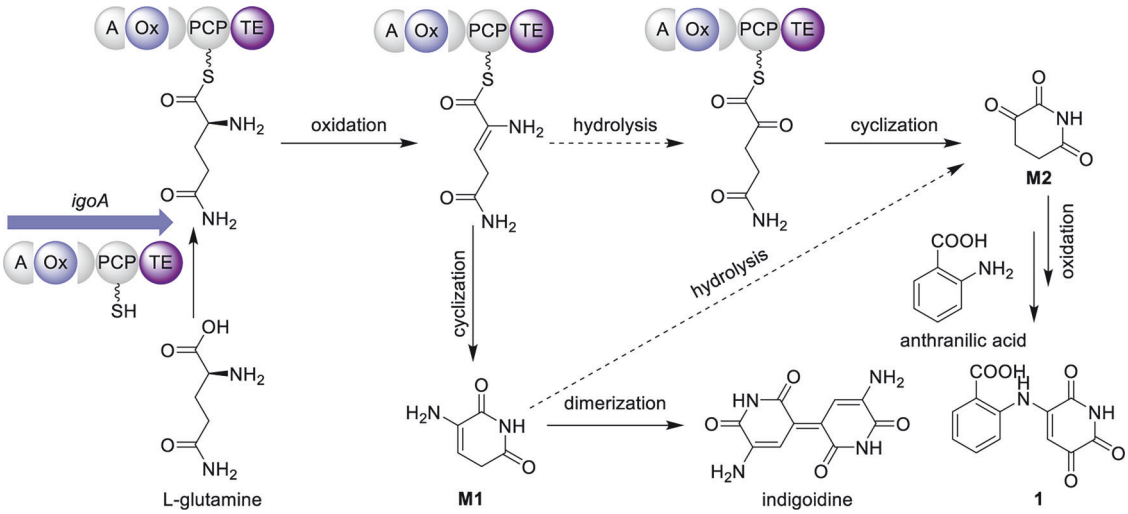

Fig. 2 Proposed biosynthetic pathway of **1** and indigoidine

including oxidative modifications and nucleophilic attacks. The inherent reactivity of the amine group of anthranilic acid is proposed to act as a nucleophile toward carbonyl groups of ketones within the growing intermediate, with these reactions ultimately being followed by dehydration steps to furnish the final complex structure of indigomycin. Taken together, we postulate a divergent biosynthesis for indigoidine and indigomycin based on bioinformatic and chemical logic, the definitive confirmation of which still requires further enzymological and genetic investigation.

In summary, this study successfully identified, cloned, and heterologously expressed the indigoidine BGC (*igo*) from *S. laurentii*, leading to the discovery of the novel pyridinetrione analogue, indigomycin. Furthermore, we propose a plausible biosynthetic model where the core NRPS (IgoA) diverges from a common dehydroglutaminyl intermediate to produce both the classic pigment indigoidine and the novel molecule of indigomycin. Notably, indigomycin is characterized as the first example featuring a pyridinetrione scaffold incorporating an anthranilic acid moiety. Collectively, this work validates the continued value of indigoidine synthetases as effective genetic markers for natural product discovery and provides a framework for systematic exploration of chemical diversity encoded within microbial genomes.

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Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

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